

NANTUCKET BUOY 44008, USA

WMO: 992790

Lat: 40.504N

Lon: 69.248W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (W05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.7	-3.8	-11.1	1.5	-3.7	-9.4	1.7	-2.8	18.1	1.3	17.3	1.6	10.3	320	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.1	24.1	23.0	23.4	22.4	22.8	21.8	23.7	23.9	23.0	23.4	22.3	22.8	5.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	18.4	23.8	22.9	17.6	23.3	22.1	16.8	22.7	70.7	24.1	68.1	23.4	65.5	22.9	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.9	13.6	-6.9	25.0	2.2	0.9	-8.5	25.7	-9.8	26.2	-11.1	26.7	-12.7	27.3		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	4.1	3.7	4.7	7.3	10.6	14.3	18.7	20.1	18.5	14.4	10.6	7.5
	DBStd	6.32	4.16	3.43	2.62	1.87	2.00	2.33	2.07	2.01	2.25	2.62	3.23	3.52
	HDD10.0	752	186	177	166	82	17	0	0	0	0	2	31	93
	HDD18.3	2709	442	409	424	329	241	121	21	8	24	123	231	336
	CDD10.0	1211	2	1	0	2	34	131	270	312	256	139	49	16
	CDD18.3	127	0	0	0	0	0	1	32	62	30	2	0	0
	CDH23.3	58	0	0	0	0	0	0	9	33	16	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.5	8.4	8.4	7.6	6.6	5.3	4.6	4.1	4.3	5.4	7.0	7.8	8.3
Precipitation	PrecAvg	1504	141	134	132	117	103	93	100	122	125	128	144	167
	PrecMax	2167	219	208	230	198	225	206	179	460	252	265	265	231
	PrecMin	1258	82	60	60	41	33	16	44	25	33	42	48	88
	PrecStd	196	36	37	39	36	39	48	39	84	62	53	43	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	13.8	11.2	12.8	17.2	20.7	24.4	25.0	24.6	21.8	18.7	16.1
		MCWB	12.9	12.9	10.6	11.9	16.8	19.1	23.6	23.6	23.6	21.1	17.2	15.5
	2%	DB	12.0	10.9	10.2	11.7	15.5	19.7	23.4	24.2	23.9	20.3	16.9	14.7
		MCWB	11.2	10.3	9.7	11.0	15.2	19.0	22.4	23.0	22.9	19.4	15.8	13.8
	5%	DB	11.0	9.8	9.5	11.0	14.4	18.8	22.6	23.7	22.9	19.0	15.9	13.8
		MCWB	10.2	9.1	8.9	10.4	13.9	18.2	21.6	22.6	22.1	17.9	14.8	12.8
	10%	DB	9.8	8.5	8.5	10.2	13.5	17.9	21.8	23.0	22.1	18.0	15.0	12.8
		MCWB	9.0	7.7	7.7	9.5	12.9	17.4	21.0	22.0	20.9	16.8	14.0	11.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	11.3	11.1	12.3	17.1	19.7	24.0	24.3	24.6	20.8	17.9	16.1
		MCDB	13.6	11.5	11.2	12.6	17.4	19.9	24.2	24.6	24.6	20.8	18.4	16.2
	2%	WB	12.1	9.9	10.1	11.3	15.6	18.9	23.1	23.8	23.5	19.4	16.7	14.4
		MCDB	12.4	10.3	10.3	11.6	15.7	19.2	23.4	24.1	23.7	19.7	17.4	14.7
	5%	WB	10.8	8.7	9.3	10.7	14.4	18.2	22.2	23.2	22.4	18.5	15.5	13.4
		MCDB	11.3	9.2	9.6	10.9	14.5	18.6	22.5	23.6	23.0	18.8	16.0	14.0
	10%	WB	9.9	7.4	8.2	9.9	13.3	17.4	21.5	22.7	21.5	17.4	14.6	12.5
		MCDB	10.4	8.1	8.7	10.2	13.5	17.8	21.9	23.1	22.2	18.0	15.2	13.2
Mean Daily Temperature Range	5% DB	MDBR	4.7	4.6	3.6	2.7	2.7	2.8	3.2	3.1	2.7	2.9	3.4	4.5
		MCDBR	6.1	6.1	4.9	4.0	3.6	3.4	3.6	3.4	3.7	3.9	4.4	5.6
		MCWBR	6.6	6.5	5.3	4.0	3.3	3.2	3.3	3.1	3.7	4.6	5.0	6.5
	5% WB	MCDBR	5.7	5.9	4.7	3.9	3.4	3.4	3.4	3.1	3.7	3.5	4.1	5.7
		MCWBR	6.6	6.5	5.4	4.1	3.3	3.2	3.3	3.0	3.8	4.4	4.7	6.5
Clear-Sky Solar Irradiance	taub	0.340	0.347	0.359	0.389	0.424	0.455	0.480	0.443	0.383	0.372	0.342	0.332	
	taud	2.433	2.419	2.426	2.358	2.297	2.248	2.196	2.303	2.433	2.462	2.511	2.487	
	Ebn at Noon	813	864	893	884	857	828	802	824	857	826	809	795	
	Edh at Noon	81	94	104	119	130	137	143	125	103	90	75	72	
All-Sky Solar Radiation	RadAvg	1.58	2.51	3.77	4.95	5.62	6.15	6.15	5.65	4.64	3.18	1.96	1.39	
	RadStd	0.14	0.24	0.33	0.49	0.57	0.49	0.35	0.39	0.39	0.23	0.22	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86	N/A

Nomenclature: See separate page